FRESHBITE – ONLINE FOOD ORDERING AND DELIVERY SYSTEM

PROJECT REPORT

1. INTRODUCTION

FreshBite is a web-based food ordering and delivery management system designed to provide users with a simple and convenient way to explore food items, select their favorite dishes, place orders, and manage their food purchases online.

Traditional food ordering methods require customers to visit restaurants or make phone calls to place orders. These methods can be time-consuming and may not provide sufficient information about available food items, prices, or order status. FreshBite addresses these problems by providing an online platform where customers can view food items and place orders from anywhere.

The system provides an easy-to-use interface for customers and helps restaurants manage food items and customer orders efficiently. The project demonstrates the practical application of web development concepts such as user interfaces, database management, form handling, authentication, and order processing.

FreshBite aims to improve the overall food ordering experience by making the process faster, easier, and more organized.

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2. ABSTRACT

FreshBite is an online food ordering system developed to simplify the process of ordering food through a digital platform. The system allows users to browse different food items, view their details and prices, add selected items to their cart, and place orders.

The application can maintain customer and food-related information and can manage orders in an organized manner. The system reduces manual work involved in traditional food ordering and provides a convenient solution for customers.

The main objective of FreshBite is to develop a user-friendly food ordering platform that connects customers with available food services. It provides a structured approach for managing food items, customers, and orders.

The project also demonstrates the use of modern software development concepts including frontend design, backend processing, database connectivity, validation, and responsive user interfaces.

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3. PROBLEM STATEMENT

In traditional food ordering systems, customers may need to visit a restaurant or contact the restaurant through telephone calls. This process can create several difficulties.

Customers may not know the complete list of available food items, updated prices, or availability. Restaurants also need to manually maintain customer orders, which can result in errors and delays.

The major problems in the traditional system include:

- Manual order processing.
- Difficulty in maintaining customer information.
- Lack of centralized food information.
- Possibility of mistakes while taking orders.
- Time-consuming ordering process.
- Difficulty in tracking orders.
- Limited accessibility for customers.

FreshBite is proposed to overcome these problems by providing a digital food ordering platform.

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4. OBJECTIVES

The major objectives of the FreshBite project are:

1. To develop an easy-to-use online food ordering system.
2. To allow users to browse available food items.
3. To display food names, descriptions, and prices.
4. To allow customers to select and order food items.
5. To maintain customer and order information.
6. To reduce manual work involved in food ordering.
7. To improve the speed and accuracy of order processing.
8. To provide a convenient interface for customers.
9. To organize food and order information efficiently.
10. To demonstrate practical web application development.

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5. EXISTING SYSTEM

In the existing traditional system, customers generally place food orders by visiting restaurants or contacting restaurants through phone calls.

The restaurant staff manually receives the order and records the customer requirements. This process can lead to communication errors and requires additional manual effort.

Disadvantages of Existing System

- Manual order management.
- More chances of human errors.

- Time-consuming process.
- Difficult to maintain records.
- Customers have limited information about food items.
- Difficult to manage large numbers of orders.
- Order tracking may not be available.
- Customer convenience is limited.

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6. PROPOSED SYSTEM

FreshBite provides a digital solution for food ordering. Users can access the application through a web interface and browse the available food items.

The customer can select the required food items and provide the necessary order information. The system processes the order and stores the information for future reference.

The proposed system reduces manual effort and improves the organization of customer and order information.

Advantages of Proposed System

- Easy online food ordering.
- Simple and user-friendly interface.
- Faster order processing.
- Better data management.
- Reduced manual errors.
- Easy management of food items.
- Convenient access for customers.
- Organized order records.
- Improved customer experience.

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7. SYSTEM REQUIREMENTS

Hardware Requirements

Component	Requirement
Processor	Intel Core i3 or higher
RAM	Minimum 4 GB

Component	Requirement
Storage	Minimum 20 GB free space
Display	Standard monitor
Keyboard	Standard keyboard
Mouse	Standard mouse
Internet	Required for online functionality

Software Requirements

Software	Purpose
Windows 10/11	Operating System
Visual Studio Code	Development Environment
HTML	Web Page Structure
CSS	Web Page Styling
JavaScript	Client-side functionality
Backend Technology	Server-side processing
Database	Data storage
Web Browser	Application execution

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8. SYSTEM DESIGN

The FreshBite system follows a structured architecture consisting of different components.

Main Components

1. User Interface

The user interface allows customers to interact with the application. It contains pages for food items, login, registration, cart, and orders.

2. Application/Backend Layer

The backend handles user requests, processes orders, validates data, and communicates with the database.

3. Database Layer

The database stores information related to users, food items, orders, and other application data.

Basic Architecture

User → Frontend → Backend → Database

The user interacts with the frontend. The frontend sends requests to the backend. The backend processes the request and communicates with the database. The result is then returned to the user interface.

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9. MODULES OF FRESHBITE

9.1 User Registration Module

This module allows new users to create an account by providing required details.

The registration information is stored in the database and can be used for authentication.

9.2 Login Module

Registered users can log in using their credentials. The system validates the entered information and provides access to the application.

9.3 Food Items Module

This module displays the food items available in the system. Users can view food names, descriptions, categories, and prices.

9.4 Search/Browse Module

Users can browse or search for food items based on their requirements. This makes it easier to find a particular item.

9.5 Cart Module

The cart allows users to add selected food items before placing an order. Users can review the selected items and their quantities.

9.6 Order Module

This module processes customer orders. After selecting the required food items, the customer can submit the order.

9.7 Order Management Module

The system maintains information about placed orders. Order details can include customer information, food items, quantity, price, and order status.

9.8 Admin Module

The administrator can manage food items and monitor customer orders. This helps maintain the application efficiently.

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10. DATABASE DESIGN

The database is an important part of the FreshBite system. It stores and manages application data.

Main Entities

User

- User_ID
- Name
- Email
- Password
- Phone
- Address

Food

- Food_ID
- Food_Name
- Category
- Description
- Price
- Image

Order

- Order_ID
- User_ID
- Order_Date
- Total_Amount
- Status

Order_Item

- Order_Item_ID
- Order_ID
- Food_ID
- Quantity
- Price

The relationships between these entities allow the system to maintain customer orders and food information effectively.

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11. IMPLEMENTATION

The FreshBite application is implemented using web development technologies.

Frontend

HTML is used to create the basic structure of web pages. CSS is used to design the interface and provide a visually attractive layout. JavaScript can be used to provide interactive features and client-side validation.

The frontend provides pages such as:

- Home Page
- Login Page
- Registration Page
- Food Menu
- Food Details
- Cart
- Order Page
- Admin Page

Backend

The backend handles application logic and communication between the frontend and database.

It processes user requests, validates information, manages orders, and retrieves required data.

Database

The database stores user information, food details, and order information. Database operations include inserting, updating, deleting, and retrieving records.

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12. WORKING OF THE SYSTEM

The working process of FreshBite can be explained as follows:

Step 1: User Registration

A new customer creates an account by entering the required information.

Step 2: User Login

The registered customer logs into the FreshBite application.

Step 3: Browse Food

The customer views the available food items.

Step 4: Select Food

The customer selects the required food items and specifies the quantity.

Step 5: Add to Cart

Selected items are added to the shopping cart.

Step 6: Review Order

The customer checks the selected items and total amount.

Step 7: Place Order

The customer confirms the order.

Step 8: Store Order

The system stores the order details in the database.

Step 9: Order Management

The restaurant/admin can view and manage the received order.

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13. TESTING

Testing is performed to verify that the FreshBite application works correctly.

Functional Testing

Functional testing checks whether each feature performs its intended operation.

Test Case	Expected Result
User Registration	Account should be created
User Login	User should successfully log in
View Food	Food items should be displayed
Add to Cart	Selected item should appear in cart
Remove Item	Item should be removed
Place Order	Order should be created
Invalid Login	Error message should be displayed
Empty Form	Validation message should appear

Testing helps identify errors and improves the reliability of the application.

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14. ADVANTAGES

FreshBite provides several benefits:

- Simple and convenient food ordering.
- Saves time for customers.
- Reduces manual order processing.
- Provides organized food information.
- Helps maintain customer records.
- Helps manage orders efficiently.
- Reduces communication errors.
- Provides better accessibility.
- Improves overall customer experience.
- Can be extended with additional features.

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15. LIMITATIONS

Although FreshBite provides an efficient food ordering solution, some limitations may exist.

- Internet connection may be required.
- Online payment functionality may require additional integration.
- Real-time delivery tracking may not be available in the basic version.
- System performance depends on server and database configuration.
- Security needs to be improved for production-level deployment.
- Notification services may require external APIs.

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16. FUTURE SCOPE

FreshBite can be enhanced with many advanced features in the future.

Online Payment

Payment gateways can be integrated to allow customers to pay online.

Live Order Tracking

GPS-based tracking can be added to show the current delivery location.

Mobile Application

A dedicated Android and iOS application can be developed.

Notifications

SMS, email, and push notifications can be implemented for order updates.

Ratings and Reviews

Customers can rate food items and provide reviews.

AI-Based Recommendations

Artificial Intelligence can be used to recommend food based on customer preferences and previous orders.

Offers and Discounts

The system can provide coupons, promotional offers, and discounts.

Advanced Admin Dashboard

An analytics dashboard can display sales, popular food items, customer activity, and order statistics.

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17. CONCLUSION

FreshBite is an online food ordering and delivery management system developed to simplify the process of ordering food. The system provides customers with an easy platform to browse food items, select products, manage their cart, and place orders.

The project reduces the dependency on traditional manual food ordering methods and provides an organized approach to managing users, food items, and orders.

Through the development of FreshBite, important concepts of web application development, database management, frontend design, backend processing, validation, and system testing can be practically understood.

The system can be further improved by integrating online payments, live delivery tracking, mobile applications, notifications, reviews, and AI-based food recommendations.

Overall, FreshBite provides a practical and user-friendly solution for digital food ordering and can be extended into a complete commercial food delivery platform.

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18. REFERENCES

1. HTML and CSS documentation.
2. JavaScript documentation.
3. Database management system documentation.
4. Web application development resources.
5. Software engineering and system design concepts.
6. Official documentation of the technologies used in the project.

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19. PROJECT SUMMARY

Project Name: FreshBite
Project Type: Web-Based Food Ordering System
Domain: Food Technology / E-Commerce
Primary Users: Customers and Administrators
Main Functions: Food Browsing, Cart Management, Ordering and Order Management
Database: Used for storing user, food and order information
Development Environment: Visual Studio Code
Platform: Web Application